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Chihuahua or Muffin

In this workshop I have learned the basic concepts of image classification using deep learning with PyTorch. My objective was to build and train a neural network that could distinguish between images of Chihuahuas and muffins. I know that these two distinct classes may seem very different, many images in the dataset are visually similar, making this a challenging classification problem. During the workshop, I learned how to prepare image data, define a neural network, train the model, and evaluate its performance on unseen validation images and also I try to improve the model and go closer to 100 as possible.

From a few techniques that I used, one of the main techniques was image preprocessing. Before training the model, since I needed to decide the image size, I resized all images to a fixed size of 300×300 pixels and converted into tensors so they could be processed by PyTorch. The images were also normalized to improve the stability of the training process. After preprocessing, the images were organized into training and validation datasets using PyTorch's `ImageFolder` and `DataLoader` classes. This made it easier to load the data in batches and train the model more efficiently. Another concept that I learned was how a neural network performs image classification. In this project, I built a fully connected neural network called `MySkynet`. The network contained several hidden layers with ReLU activation functions and an output layer that predicted whether an image was a Chihuahua or a muffin. I also learned how the forward pass, loss function, backpropagation, and optimizer work together to update the model's weights during training.

During this activity, I also became more familiar with several important machine learning concepts, including image classification, neural networks, loss functions, optimizers, epochs, learning rate, batch size, and validation accuracy. Like I already mentioned in previous homework's, I did a class about machine learning, where I learned some of this material, but this time I used on a image dataset. Even though this project did not use a convolutional neural network or transfer learning, I learned why these techniques are commonly used in image classification. CNNs are designed to recognize important visual features such as edges, textures, and shapes, making them much more effective than simple fully connected networks for image recognition tasks. Transfer learning allows developers to use pre-trained models that have

already learned useful image features, reducing both training time and the amount of data required.

One of the biggest challenges I faced was getting the notebook to run correctly. I mean, I started this homework a few times, last time I spent more than 4 hours trying to make the model work better, and all time I was getting a different result. I mean this results are not even good results, any parameter that I change it, always killed the results. But maybe that was on me because today I made it work. This process taught me the importance of debugging and understanding each part of the code. Another challenge was understanding how different hyperparameters affect the training process. Choosing values such as the batch size, learning rate, and number of epochs required some experimentation. I learned that these parameters directly influence how quickly and accurately the model learns. Although my first model was not perfect, it successfully classified most of the validation images correctly. The few mistakes usually occurred on images where muffins looked very similar to Chihuahuas or vice versa, showing how difficult image classification can be even for humans.

This project also helped me better understand the complete machine learning workflow. I learned that building a model is only one part of the process. Like my previous teacher said one day, A good clean data is a sign of a good model. Preparing the data correctly, selecting appropriate parameters, training the model, validating its performance, and analyzing prediction results are all equally important steps. Looking at the prediction probabilities for each validation image allowed me to understand not only whether the model was correct but also how confident it was in each prediction.

The techniques learned in this workshop have many practical applications. Image classification is used in healthcare to detect diseases from medical images, in manufacturing to identify defective products during quality inspection, in agriculture to recognize plant diseases, in transportation for traffic sign recognition in self-driving vehicles, and for last but not least, in the security systems that use facial recognition, to put criminals behind the bars. Many businesses also use image classification to automatically organize photos, identify products, and improve customer experiences.

Overall, I found this workshop both challenging and rewarding. It required patience because I had to solve some coding errors before the model could train successfully. However, fixing these problems helped me better understand how each part of the neural network works. Seeing the model correctly classify most of the Chihuahua and muffin images was satisfying because it showed that the training process was successful. This experience increased my confidence in working with PyTorch and deep learning, and it motivated me to continue learning more advanced techniques such as convolutional neural networks and transfer learning. I believe the knowledge gained from this workshop will be valuable for future projects on this class.

